6.2 Cloning and biotechnology

6.2.1 Cloning and biotechnology

Farmers and growers exploit "natural" vegetative propagation in the production of uniform crops. Artificial clones of plants and animals can now be produced.

Biotechnology is the industrial use of living organisms (or parts of living organisms) to produce food, drugs or other product.

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	**(i)** natural clones in plants and the production of natural clones for use in horticulture **(ii)** how to take plant cuttings as an example of a simple cloning technique	To include examples of natural cloning and the methods used to produce clones (various forms of vegetative propagation). Dissection of a selection of plant material to produce cuttings. **PAG2** HSW4
(b)	**(i)** the production of artificial clones of plants by micropropagation and tissue culture **(ii)** the arguments for and against artificial cloning in plants	To include an evaluation of the uses of plant cloning in horticulture and agriculture. HSW9, HSW12
(c)	natural clones in animal species	To include examples of natural clones (twins formed by embryo splitting).

(d) **(i)** how artificial clones in animals can be produced by artificial embryo twinning or by enucleation and somatic cell nuclear transfer (SCNT) **(ii)** the arguments for and against artificial cloning in animals	To include an evaluation of the uses of animal cloning (examples including in agriculture and medicine, and issues of longevity of cloned animals). HSW9, HSW10, HSW12
(e) the use of microorganisms in biotechnological processes	To include reasons why microorganisms are used; economic considerations, short life cycle and growth requirements.
(f) the advantages and disadvantages of using microorganisms to make food for human consumption	To include bacterial and fungal sources. HSW9, HSW12
(g) **(i)** how to culture microorganisms effectively, using aseptic techniques **(ii)** the importance of manipulating the growing conditions in batch and continuous fermentation in order to maximise the yield of product required	An opportunity for serial dilutions and culturing on agar plates. **PAG7** HSW4
(h) **(i)** the standard growth curve of a microorganism in a closed culture	To include the formula for number of individual organisms $N = N_0 \times 2^n$
(ii) practical investigations into the factors affecting the growth of microorganisms	An opportunity for serial dilutions and the use of broth. *M0.1, M0.3, M0.5, M1.1, M1.3, M2.5, M3.1, M3.2, M3.4, M3.5, M3.6* **PAG7** HSW4

(i)	the uses of immobilised enzymes in biotechnology and the different methods of immobilisation.	To include methods of enzyme immobilisation **AND** an evaluation of the use of immobilised enzymes in biotechnology examples could include: • glucose isomerase for the conversion of glucose to fructose • penicillin acylase for the formation of semi-synthetic penicillins (to which some penicillin-resistant organisms are not resistant) • lactase for the hydrolysis of lactose to glucose and galactose • aminoacylase for production of pure samples of L-amino acids • glucoamylase for the conversion of dextrins to glucose. Learners are **not** required to recall the examples above but will be expected to apply their knowledge and understanding of immobilised enzymes in the context of biotechnology. *M0.2, M0.3, M1.2, M1.3, M1.4, M1.6, M1.10, M3.2, M4.1* **PAG4** HSW4